Module 5 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 5, Quiz B: Concept check

Reading: Module 5 Read 2 (complete Read 1 + Quiz A first)
Canvas: Concept Check quiz (Module 5)

Apply residual conditions for inference, interpret CIs for slope, and distinguish CI for mean response from prediction interval.

Full regression workflow, R output, and lab situations → Synthesis.

Section A — Residual conditions for inference (Read 2 §1)

A1. Normality condition

What assumption about residuals supports confidence and prediction intervals in regression?

A2. Checking residuals

Name two visual checks used to assess residual normality in regression.

A3. When inference is unreliable

If a residual plot shows a clear funnel shape (spread increases with x), what concern does this raise?

Answer key — Section A — Residual conditions for inference (Read 2 §1)

1. Residuals should be approximately **normally distributed** with **constant spread** (homoscedasticity) across all values of x.
2. A **histogram of residuals** (should look roughly bell-shaped) and a **Q-Q plot** (points should fall near the diagonal line).
3. Non-constant variance (**heteroscedasticity**) — the equal-spread condition is violated, so standard CIs and p-values may be unreliable.

Section B — CI for slope (Read 2 §1)

B1. CI for slope interpretation

R output: `confint(fit, 'study_hours', level = 0.95)` → `[2.44, 3.68]`. Write the one-sentence interpretation.

B2. Significance from CI

The 95% CI for slope β_1 is $[1.1, 3.7]$. Is there evidence that $\beta_1 \neq 0$? Explain.

B3. No evidence of association

The 95% CI for a slope is $[-0.3, 1.8]$. Interpret this result.

Answer key — Section B — CI for slope (Read 2 §1)

1. We are 95% confident that each additional hour of study changes the **mean** exam score by between **2.44 and 3.68 points**.
2. **Yes** — the CI does not contain 0, so there is evidence that study hours has a real (non-zero) linear effect on exam score.
3. The CI contains 0, so there is **no convincing evidence** of a linear association; zero slope is consistent with the data.

Section C — CI for mean response vs prediction interval (Read 2 §1)

C1. Mean response CI interpretation

At $x_0 = 10$ study hours, `predict(fit, interval = 'confidence')` $\to$ `[76.1, 80.5]`. Interpret.

C2. Prediction interval interpretation

At $x_0 = 10$ study hours, `predict(fit, interval = 'prediction')` $\to$ `[62.3, 94.3]`. Interpret.

C3. Why PI is wider

Explain in one sentence why the prediction interval is wider than the confidence interval for the mean response.

C4. Which interval to use

A student wants to know what **score they personally** can expect to earn. Should they use the CI for the mean or the prediction interval? Why?

C5. R syntax for prediction interval

Write the R call to compute a 95% prediction interval for a new observation at `dose = 7` using a fitted model `fit_plant`.

Answer key — Section C — CI for mean response vs prediction interval (Read 2 §1)

1. We are 95% confident the **mean** exam score among all students who study 10 hours is between **76.1 and 80.5 points**.
2. We predict with 95% confidence that **one individual** student who studies 10 hours will score between **62.3 and 94.3 points**.
3. The prediction interval must account for both uncertainty in the **mean** (same as the CI) **plus** the individual's random deviation from that mean.
4. The **prediction interval** — it covers where an individual future observation is likely to fall, not just where the average falls.
5. `predict(fit_plant, newdata = data.frame(dose = 7), interval = "prediction", level = 0.95)`
